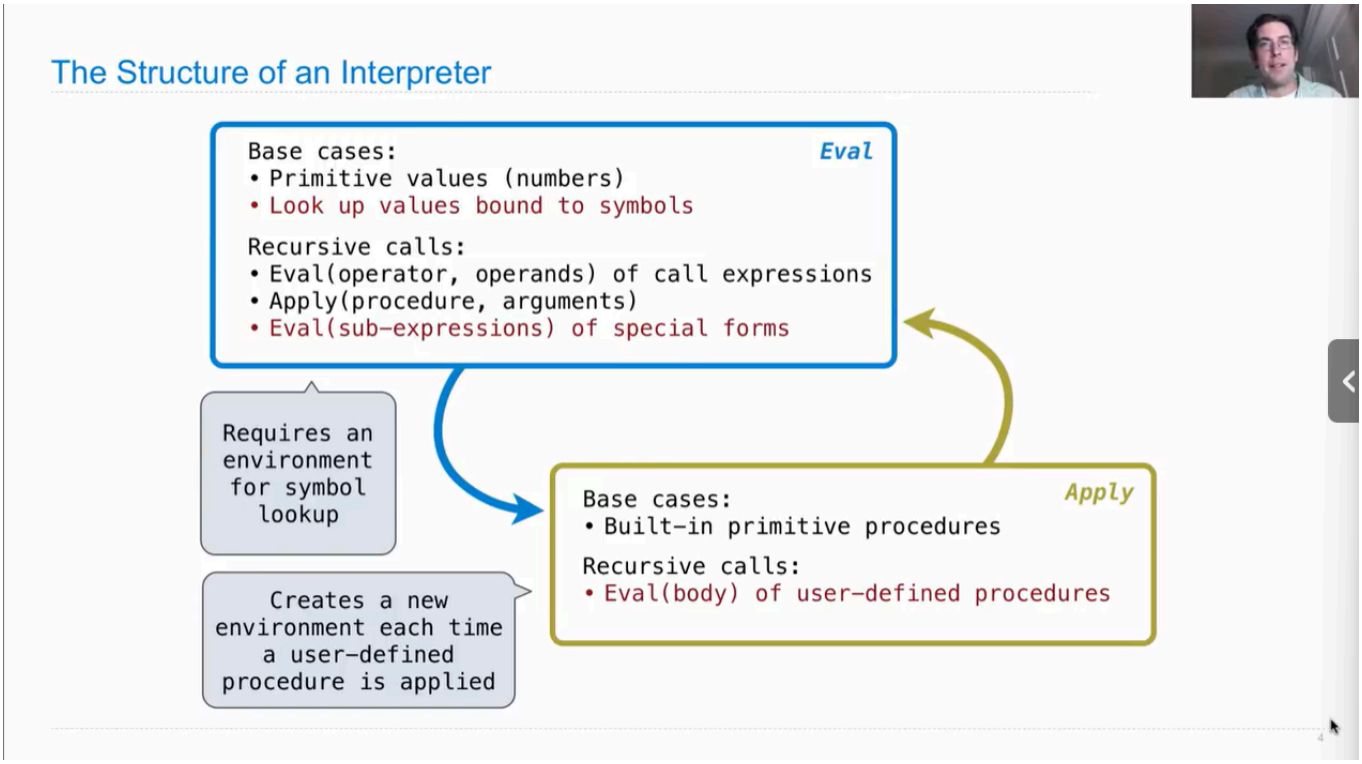
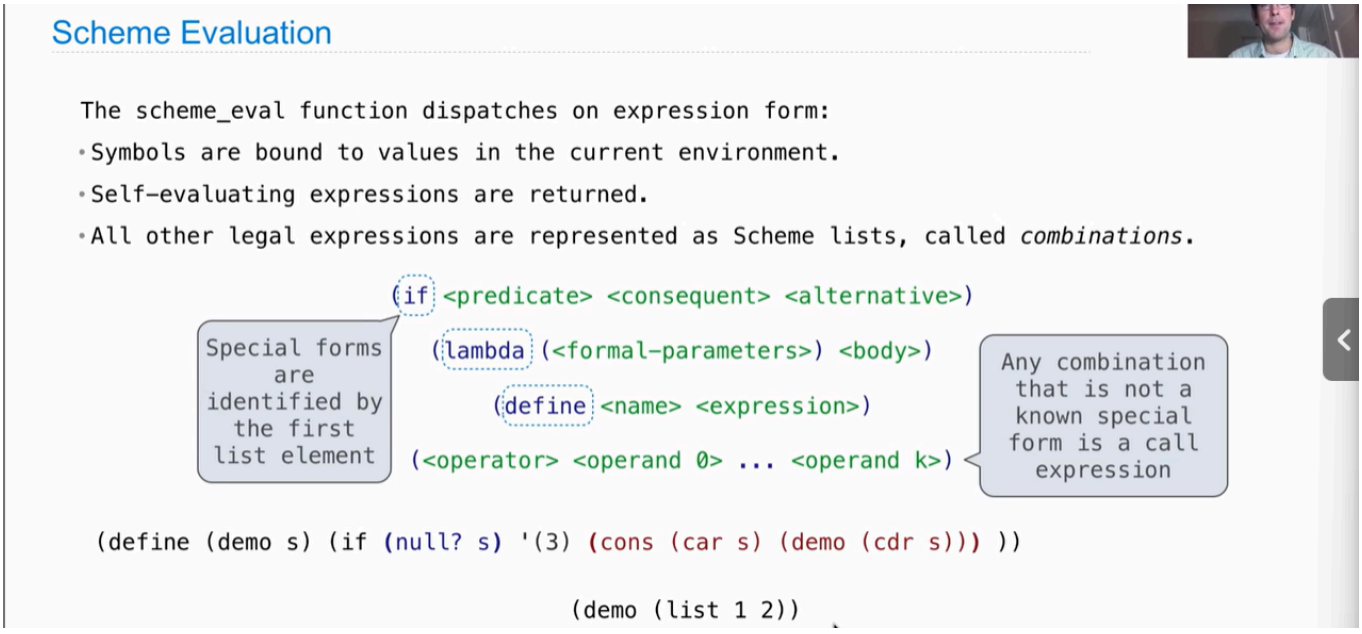
Lecture 30.Interperters

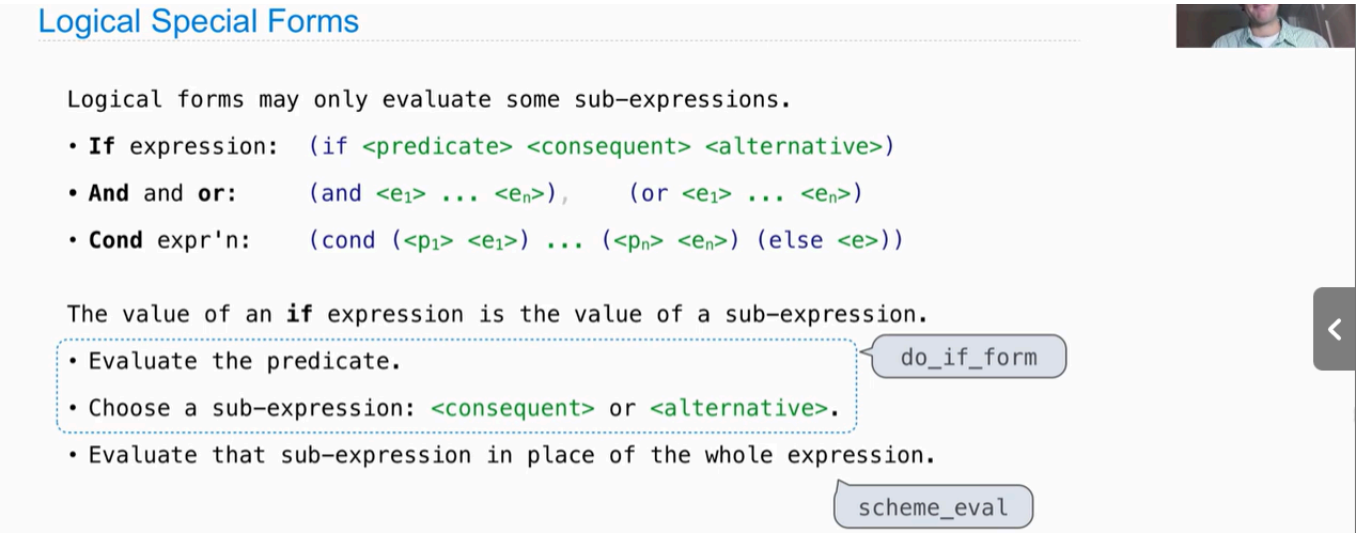
Interpreting Scheme



Special Forms

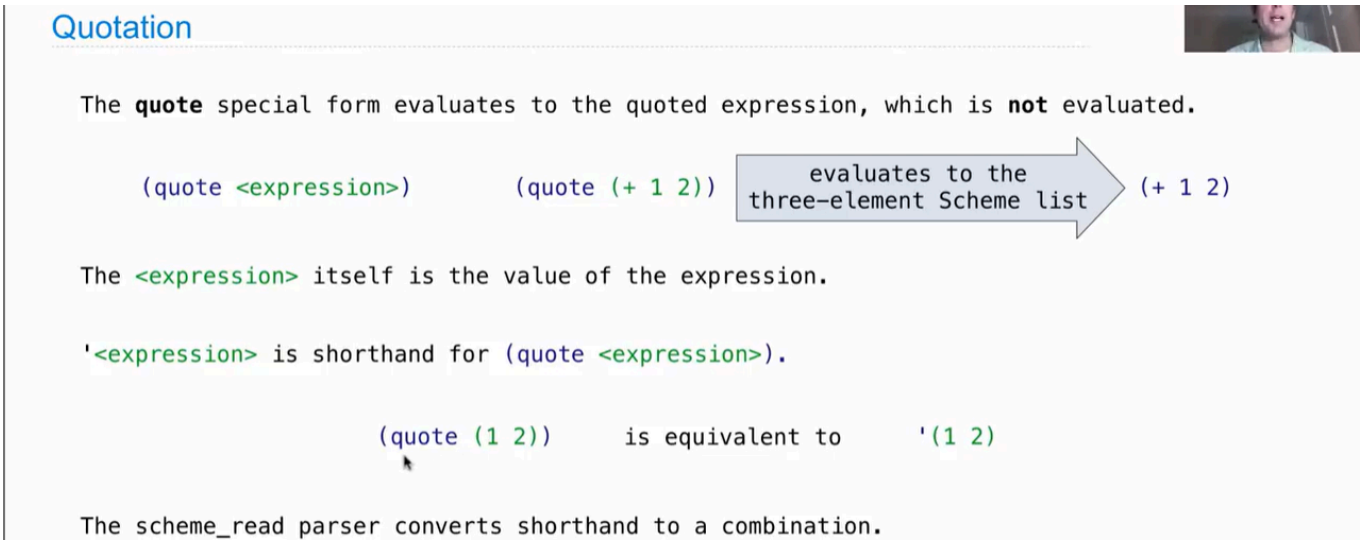


Logical Special Forms

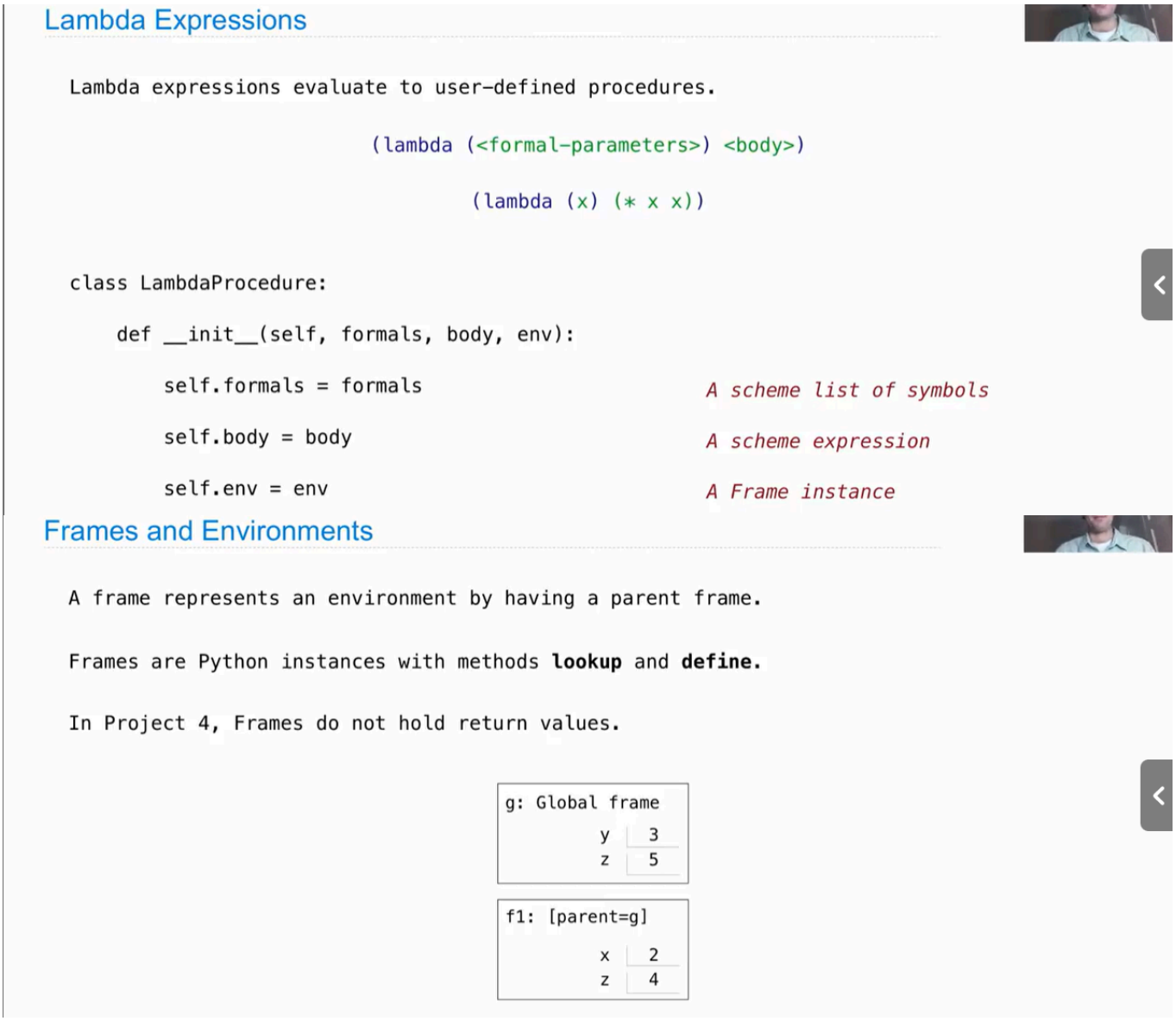


> I'm really curious about how this thing is implemented, since I have no idea how to organize all these things.

Quotation



Lambda Expressions



we use the define method to bind values to the frame and add variables. then we use lookup to see the value bind to the variable in the special frame.

Define Expressions

Define Expressions

Define binds a symbol to a value in the first frame of the current environment.

```
(define <name> <expression>)
```

1. Evaluate the <expression>.
2. Bind <name> to its value in the current frame.

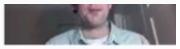
```
(define x (+ 1 2))
```

Procedure definition is shorthand of define with a lambda expression.

```
(define (<name> <formal parameters>) <body>)
```

```
(define <name> (lambda (<formal parameters>) <body>))
```

Applying User-Defined Procedures



To apply a user-defined procedure, create a new frame in which formal parameters are bound to argument values, whose parent is the **env** of the procedure.

Evaluate the body of the procedure in the environment that starts with this new frame.

```
(define (demo s) (if (null? s) '(3) (cons (car s) (demo (cdr s)))))
```

```
(demo (list 1 2))
```